

SAUMYA AGRAHARI

Software Development Engineer | Full Stack Developer | AI/ML Engineer | Python Developer

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Elsevier Published Researcher | **3x Hackathon Winner** (UHack 4.0 #1/500+ Participants, Google Build with AI GDG, NSUT Top 10) | **93.78% ML Accuracy** | **CGPA: 8.4** | **SGPA: 9.0**

Professional Summary

Final-year B.Tech (AI/ML) student with 2+ years of experience — Elsevier Procedia published researcher, 2 production internships, 3 hackathon wins, 5+ deployed full-stack and ML applications. Proven in building scalable REST APIs, ML pipelines, and full-stack systems using Python, React.js, Node.js, Django, TensorFlow, Docker, and CI/CD pipelines. Strong foundation in System Design, Agile/Scrum, and cloud deployment (AWS, GCP, Vercel). Available for SDE / Full Stack / AI-ML Engineer roles (PPO/internship 2026; full-time May 2027).

Education

B.Tech — Artificial Intelligence & Machine Learning

United University, Prayagraj

Aug 2023 – May 2027

CGPA: 8.4/10 | SGPA: 9.0

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, DSA, DBMS, Operating Systems, Computer Networks, System Design, Agile Software Development

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C

Frontend: React.js, Next.js, Redux, Context API, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive Design

Backend: Node.js, Django, Flask, REST API Design, JWT Authentication, API Integration, Microservices

Databases: MySQL, PostgreSQL, Firebase Firestore, Database Design, Query Optimization

AI/ML: TensorFlow, Scikit-learn, XGBoost, OpenCV, MediaPipe, Computer Vision, NLP, NumPy, Pandas, Feature Engineering, Model Deployment

Generative AI: Google Gemini API, LLM Integration, Prompt Engineering, Vertex AI

Cloud & DevOps: AWS (S3, EC2), GCP, GitHub Actions (CI/CD), Docker, Git, Vercel, Linux

Testing: Jest, Pytest, Unit Testing, API Testing, Test-Driven Development (TDD)

Core CS: DSA, OOP, System Design, SDLC, Agile/Scrum, Software Architecture

Work Experience

Prodesk IT, Noida

Backend Development Intern

Aug 2025 – Present

- Engineered 10+ REST API endpoints in Python/Django for R&D production applications, improving data retrieval by 25% via query optimization and indexing strategies.
- Wrote Pytest unit tests for 3 core API modules, achieving 85%+ test coverage; designed scalable MySQL database workflows reducing backend response time by 30%.
- Collaborated in Agile/Scrum team of 6+ engineers across 3 production releases; participated in daily standups, sprint planning, and code reviews.

Code Resite, Prayagraj

Web Developer Intern (Contract)

Jun 2025 – Jul 2025

- Delivered full-stack features across 5+ client projects using React.js, JavaScript, Node.js, and REST APIs; reduced UI bugs by 40% via systematic testing and debugging.
- Improved page load performance by 35% through frontend optimization (lazy loading, code splitting, caching); all projects deployed on Vercel with CI/CD pipelines.

Hanuman Pushpavarsha Committee, Prayagraj

Freelance Full Stack Developer

Jan 2024 – Mar 2025

- Built bilingual (Hindi-English) production website using Next.js, Firebase, and Vercel with 6 modules: Razorpay payment gateway, live streaming, member registry, admin dashboard; serving 500+ users.
- Reduced admin overhead by 60% by automating member registration via Firebase Firestore real-time database; implemented Firebase Auth with role-based access control (RBAC).

Projects

SignSetu — AI Bidirectional Sign Language Platform | *Python, TensorFlow, OpenCV, MediaPipe, React.js, Flask, Firebase, Docker*
[Live](#) | [GitHub](#)

- Winner, UHack 4.0** (1st/500+ participants): real-time bidirectional ISL communication platform for the deaf and hard-of-hearing community; deployed with Docker and Vercel CI/CD.
- Trained TensorFlow CNN + MediaPipe achieving **91%+ gesture recognition at 24 fps**; curated 5,000+ sample custom ISL dataset from scratch using feature engineering and data augmentation.
- Architected Sign-to-Text, Sign-to-Speech, and Text-to-Sign (3D avatar) pipelines with JWT authentication, Firebase backend, and REST API endpoints.

QuickSeva — AI Civic Services Assistant | *Python, React.js, Google Gemini API, LLM Integration, Prompt Engineering, Firebase* [Live](#) | [GitHub](#)

- Winner, Google Build with AI Hackathon** (GDG): LLM-powered assistant using Google Gemini API + prompt engineering to help citizens navigate government schemes via conversational UI.
- Designed REST API backend with Firebase Firestore session management, React.js frontend with real-time streaming responses; deployed on Vercel with GitHub Actions CI/CD.

- Open-source platform: 9 therapy modules, 100+ active users; Firebase Auth, Firestore, doctor consultation referral flow, fully responsive UI with Agile development workflow.

Research Publication

Multi-Class Anomaly Detection in Network Traffic Using Supervised Machine Learning | [DOI](#)

Saumya Agrahari, Ayush Tiwari, Priyanka | Elsevier Procedia Computer Science | ICMLDE 2025 | AICCT-2025, Sept 2025

- ML-based IDS on CIC UNSW-NB15 (175,000+ records); benchmarked 4 models (Decision Tree, Random Forest, Logistic Regression, XGBoost); **93.78% accuracy with Precision, Recall, F1, ROC-AUC**. Selected for international conference presentation.

Awards & Leadership

- **Winner, UHack 4.0:** SignSetu — 1st/500+ participants, real-time ISL AI platform.
- **Winner, Google Build with AI (GDG):** QuickSeva — recognized by Google Developer Group ecosystem judges.
- **Top 10 Finalist, NSUT National Hackathon, New Delhi:** National-level competitive hackathon.
- **Lead Organizer, HackDiwas 3.0:** 500+ participant state-level hackathon; **TEDx United University Organizer:** speaker outreach and technical operations.
- **Community:** GDG Prayagraj DevFest 2025 (GenAI, Vertex AI), FOSS Prayagraj (Design & Media).